

[illegible]

$$Y_1 = 3 \binom{(4)}{K_1} + 5 \binom{(4)}{K_{15}} \text{ \& } Y_2 = 2 \binom{(4)}{K_{15}} - \binom{(4)}{K_3} \text{ \& } \text{Det}(0^+) = -\frac{1}{2} k^2 (3 \binom{(4)}{K_1} + 5 \binom{(4)}{K_{15}}) \text{ \& } \text{Det}(1^+) = \frac{3}{2} k^2 (2 \binom{(4)}{K_{15}} - \binom{(4)}{K_3})$$
$$\begin{aligned} & \frac{4}{\kappa} \frac{1}{15} \partial_\beta \mathcal{K}^\delta_{\chi\delta} \partial^\chi \mathcal{K}^\alpha{}_\beta + \frac{5}{3} \frac{4}{\kappa} \frac{1}{15} \partial_\chi \mathcal{K}^\delta{}_{\beta\delta} \partial^\chi \mathcal{K}^\alpha{}_\beta + \frac{4}{\kappa} \frac{1}{3} \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^\delta{}_{\alpha\chi} + \\ & 6 \frac{4}{\kappa} \frac{1}{15} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^\delta{}_{\beta\chi} - 2 \frac{4}{\kappa} \frac{1}{3} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^\delta{}_{\beta\chi} + 3 \frac{4}{\kappa} \frac{1}{15} \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^\delta{}_{\chi\alpha} - \frac{4}{\kappa} \frac{1}{3} \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^\delta{}_{\chi\alpha} - \\ & 4 \frac{4}{\kappa} \frac{1}{15} \partial^\chi \mathcal{K}^\alpha{}_\beta \partial_\delta \mathcal{K}^\delta{}_{\chi\beta} - \frac{1}{2} \frac{4}{\kappa} \frac{1}{3} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^\delta{}_{\beta\chi} + \frac{4}{\kappa} \frac{1}{15} \partial_\delta \mathcal{K}^{\alpha\beta\chi} \partial^\delta \mathcal{K}^{\alpha\beta\chi} + \frac{4}{\kappa} \frac{1}{15} \partial_\delta \mathcal{K}^{\alpha\beta\chi} \partial^\delta \mathcal{K}^{\alpha\beta\chi} \end{aligned}$$

A diagram showing two vertices (circles) connected by a horizontal line. Above the line is a green arrow pointing right, labeled k^μ . Each vertex has two external lines extending outwards.